

Q1. Fluid Properties + Viscosity Concepts

Typical ask: Define density/specific weight/SG/viscosity, compute one from given data, identify Newtonian vs non-Newtonian.

Fluid: deforms continuously under shear stress. Liquids keep volume; gases fill container.

Density: mass per unit volume.

$$\rho = \frac{m}{V} \quad [\text{SI: kg/m}^3; \text{US: lbm/ft}^3]$$

Specific volume: inverse of density.

$$v_s = \frac{1}{\rho} \quad [\text{SI: m}^3/\text{kg}; \text{US: ft}^3/\text{lbm}]$$

Specific weight: weight per unit volume.

$$\gamma = \frac{W}{V} = \rho g \quad [\text{SI: N/m}^3; \text{US: lbf/ft}^3]$$

Specific gravity: fluid density relative to water.

$$SG = \frac{\rho}{\rho_w} = \frac{\gamma}{\gamma_w} \quad [-]$$

Water refs: $\rho_w \approx 1000 \text{ kg/m}^3 = 62.4 \text{ lbm/ft}^3$; $\gamma_w \approx 9.81 \text{ kN/m}^3 = 62.4 \text{ lbf/ft}^3$.

Weight:

$$W = mg \quad [\text{SI: N}; \text{US: lbf}]$$

Circle area:

$$A = \frac{\pi D^2}{4} \quad [\text{SI: m}^2; \text{US: ft}^2]$$

Dynamic viscosity: resistance to shear deformation.

$$\tau = \mu \frac{du}{dy} \quad [\tau : \text{Pa or lbf/ft}^2]$$

$$\mu = \frac{\tau}{du/dy} \quad [\text{SI: Pa}\cdot\text{s}; \text{US: lbf}\cdot\text{s/ft}^2]$$

Kinematic viscosity: momentum diffusivity.

$$\nu = \frac{\mu}{\rho} \quad [\text{SI: m}^2/\text{s}; \text{US: ft}^2/\text{s}]$$

Newtonian fluid: shear stress is directly proportional to shear rate; μ stays constant. Examples: water, air, alcohol, gasoline, mineral oil.

Non-Newtonian fluid: apparent viscosity changes with shear rate or yield stress.

- **Shear-thinning:** viscosity decreases with shear rate.
- **Shear-thickening:** viscosity increases with shear rate.
- **Viscoplastic/Bingham:** behaves rigid until yield stress is exceeded.

Quick refs: $g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$; $1 \text{ atm} = 101.325 \text{ kPa} \approx 14.7 \text{ psi}$; $1 \text{ psi} = 144 \text{ psf}$; $1 \text{ hp} = 745.7 \text{ W}$; $1 \text{ hp} = 550 \text{ ft}\cdot\text{lbf/s}$.

Prefixes: micro 10^{-6} ; milli 10^{-3} ; kilo 10^3 ; mega 10^6 .

Q2. Pressure, Hydrostatics, Manometers

Typical ask: Find pressure at depth, convert gage/absolute, solve differential manometer, explain why pressure rises with depth.

Pressure: normal force per area.

$$P = \frac{F}{A} \quad [\text{SI: Pa or kPa}; \text{US: psi or psf}]$$

Absolute vs gage:

$$P_{abs} = P_{gage} + P_{atm} \quad [\text{same pressure units}]$$

Hydrostatic relation: pressure increases with depth and decreases with elevation.

$$\Delta P = \gamma h \quad [\text{SI: Pa or kPa}; \text{US: psi or psf}]$$

Pressure head: pressure expressed as fluid height.

$$h_p = \frac{P}{\gamma} \quad [\text{SI: m}; \text{US: ft}]$$

Piezometric head: convert pressure above liquid into equivalent fluid height.

$$h_A = \frac{P_A}{\gamma} \quad [\text{m or ft}] \quad h_{AC} = h_A + h_C \quad [\text{m or ft}]$$

Manometer rule: move **down** a fluid column $\Rightarrow +\gamma h$; move **up** $\Rightarrow -\gamma h$.

$$P_A + \sum (\pm \gamma h_i) = P_B \quad [\text{pressure units}]$$

Common trap: if using P/γ in US units, convert psi to psf first.

Q3. Submerged Plane Areas (must know old topic)

Typical ask: Compute resultant force and center of pressure on an inclined or vertical plane area.

Definitions: F_R resultant force, CT centroid, CP center of pressure, h_C vertical depth to centroid, L_C distance along plane to centroid, L_P distance along plane to CP.

Core equations:

$$h_C = L_C \sin \theta \quad [\text{m or ft}]$$

$$A = BH \quad [\text{m}^2 \text{ or ft}^2] \quad I_C = \frac{BH^3}{12} \quad [\text{m}^4 \text{ or ft}^4]$$

$$F_R = \gamma h_C A \quad [\text{SI: N}; \text{US: lbf}]$$

$$L_P = L_C + \frac{I_C}{L_C A} \quad [\text{m or ft}]$$

$$h_P = L_P \sin \theta = h_C + \frac{I_C \sin^2 \theta}{h_C A} \quad [\text{m or ft}]$$

Vertical wall shortcut: if top edge is at free surface,

$$F_R = \gamma b \frac{d^2}{2} \quad [\text{N or lbf}] \quad h_P = \frac{2d}{3} \quad [\text{m or ft}]$$

Step-by-step:

- Mark point where plane meets free surface.
- Locate centroid of area.
- Find L_C then $h_C = L_C \sin \theta$.
- Find A and I_C .
- Compute $F_R = \gamma h_C A$.
- Compute L_P or h_P for CP.
- CP is always below centroid.



Q4. Buoyancy + Bernoulli / Energy Equation

Typical ask: Find required holding force on submerged body, solve flow between two points, decide what heads cancel.

Buoyancy principle: buoyant force equals weight of displaced fluid.

$$F_b = \gamma_f V_d \quad [\text{SI: N}; \text{US: lbf}]$$

$$W_o = \gamma_o V_o \quad [\text{SI: N}; \text{US: lbf}]$$

Buoyancy method: draw FBD, write $\sum F_y = 0$, include F_b , weight, and external/support force.

[Flow Rate]:

$$Q = Av \quad [\text{SI: m}^3/\text{s}; \text{US: ft}^3/\text{s}]$$

[Continuity]:

$$Q_1 = Q_2 \quad A_1 v_1 = A_2 v_2$$

Energy forms:

$$PE = Wz = mgz \quad [\text{SI: J}; \text{US: ft}\cdot\text{lbf}]$$

$$KE = W \frac{v^2}{2g} = m \frac{v^2}{2} \quad [\text{SI: J}; \text{US: ft}\cdot\text{lbf}]$$

$$FE = W \frac{P}{\gamma} = m \frac{P}{\rho} \quad [\text{SI: J}; \text{US: ft}\cdot\text{lbf}]$$

[Bernoulli, ideal flow]:

$$\frac{P_1}{\gamma} + z_1 + \frac{v_1^2}{2g} = \frac{P_2}{\gamma} + z_2 + \frac{v_2^2}{2g} \quad [\text{all terms in m or ft}]$$

Bernoulli restrictions: steady, incompressible, no pump/turbine, no losses/friction, no heat transfer, along same streamline.

Heads: pressure head P/γ , elevation head z , velocity head $v^2/2g$.

Cancel when valid: open to atmosphere $\Rightarrow P_g = 0$ on both ends; large tank surface $\Rightarrow v \approx 0$; same elevation $\Rightarrow z$ cancels; same diameter and same $Q \Rightarrow v$ may match.

[General energy equation]:

$$\frac{P_1}{\gamma} + z_1 + \frac{v_1^2}{2g} + h_A - h_R - h_L = \frac{P_2}{\gamma} + z_2 + \frac{v_2^2}{2g}$$

$h_A/h_R/h_L$: h_A head added by pump; h_R head removed by motor/turbine; h_L head lost to friction/minor losses.

Pump/motor power:

$$[\text{Weight Flow}] : W_f = \gamma Q \quad [\text{SI: N/s}; \text{US: lbf/s}]$$

$$[\text{Pump Fluid Power}] : P_A = h_A \gamma Q \quad [\text{SI: W}; \text{US: ft}\cdot\text{lbf/s or hp}]$$

$$[\text{Pump Efficiency}] : \epsilon_M = \frac{P_A}{P_r} \quad [-] \quad [\text{Motor Power}] : P_R = h_R \gamma Q \quad [\text{W or hp}]$$

Q5. Reynolds Number + Major Pipe Losses

Typical ask: Find flow regime, friction factor, and major loss in circular or non-circular flow.

Reynolds number: inertia/viscous-force ratio.

$$Re = \frac{\rho v L}{\mu} = \frac{v L}{\nu} \quad [-]$$

Circular pipe:

$$Re = \frac{\rho v D}{\mu} = \frac{v D}{\nu} \quad [-]$$

Regimes: laminar $Re < 2000$; transitional $2000 < Re < 4000$; turbulent $Re > 4000$.

Major loss (Darcy-Weisbach):

$$h_L = f \frac{L}{D} \frac{v^2}{2g} \quad [\text{m or ft}]$$

Laminar flow:

$$f = \frac{64}{Re} \quad [-] \quad h_L = \frac{32 \mu L v}{\gamma D^2} \quad [\text{m or ft}]$$

Turbulent flow (explicit form):

$$f = \frac{0.25}{\left[\log_{10} \left(\frac{1}{3.7(D/\epsilon)} + \frac{5.74}{Re^{0.9}} \right) \right]^2} \quad [-]$$

Interpretation: rougher pipe $\Rightarrow$ larger f ; larger relative smoothness $D/\epsilon \Rightarrow$ smaller f . In complete turbulence, Re matters much less.

Workflow:

- Find A , then $v = Q/A$.
- Compute Re .
- Pick regime and find f .
- Compute major loss h_L .

Pipe roughness design values:

Material	ϵ m	ϵ ft
Glass	smooth	smooth
Plastic	3.0e-7	1.0e-6
Drawn tubing	1.5e-6	5.0e-6
Commercial steel	4.6e-5	1.5e-4
Galvanized iron	1.5e-4	5.0e-4
Ductile coated	1.2e-4	4.0e-4
Ductile uncoated	2.4e-4	8.0e-4
Concrete good	1.2e-4	4.0e-4
Riveted steel	1.8e-3	6.0e-3

Q6. Non-Circular Sections + Minor Losses

Typical ask: Compute hydraulic radius/diameter; use fittings and valves, combine major + minor losses.

Hydraulic radius:

$$R = \frac{A}{WP} \quad [\text{m or ft}]$$

Hydraulic diameter:

$$D_h = 4R \quad [\text{m or ft}]$$

Non-circular Reynolds number:

$$Re = \frac{\rho v (4R)}{\mu} = \frac{v (4R)}{\nu} \quad [-]$$

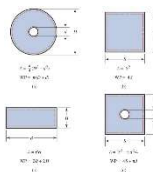
Non-circular major loss:

$$h_L = f \frac{L}{4R} \frac{v^2}{2g} \quad [\text{m or ft}]$$

Wetted perimeter: only the boundary actually touching the fluid counts. Free surface is not wetted perimeter.

Hydraulic geometry refs:

Shape	A	WP
Annulus	$\frac{\pi}{4}(D^2 - d^2)$	$\pi(D + d)$
Square	S^2	$4S$
Rectangle	BH	$2B + 2H$
Square + hole	$S^2 - \frac{\pi d^2}{4}$	$4S + \pi d$



Q7. Series + Parallel Pipe Systems

Typical ask: Find total loss in series, determine branch flow split in parallel, or solve for pressure/head/power.

Series system: same flow through every section.

$$Q_1 = Q_2 = Q_3 = \dots \quad [\text{m}^3/\text{s or ft}^3/\text{s}]$$

$$h_{L\text{total}} = \sum h_{Lj} + \sum h_{\text{minor},j} \quad [\text{m or ft}]$$

Series class idea: **Class I only:** given pipe size + $Q \Rightarrow$ find pressure/head/power.

Parallel system: flow splits and recombines.

$$Q_T = Q_A + Q_B + \dots \quad [\text{m}^3/\text{s or ft}^3/\text{s}]$$

Key condition: main line splits into two branches (like bypass / heat exchanger), then rejoins; use equal branch head loss between split and merge.

$$h_{LA} = h_{LB} = h_{LC} = \dots \quad [\text{m or ft}]$$

Procedure:

- Guess branch flow split.
- Compute v, Re, f, h_L in each branch.
- Adjust until branch losses match.
- Make sure total flow adds correctly.

Common setup: main line splits into two branches (like bypass / heat exchanger), then rejoins; use equal branch head loss between split and merge.

Q8. Pumps, Pump Curves, NPSH, Cavitation, Pneumatics

Typical ask: Choose operating point, use affinity laws, compute NPSH available, explain cavitation risk, or solve compressor/pneumatic relation.

Pump curve: head supplied by a pump as a function of flow rate.

System curve: head required by system; rises with flow because losses increase.

Operating point: pump curve $\cap$ system curve; actual Q and head delivered.

Pump type table:

Main	Subtype	Pump types
Positive displacement	Rotary	Gear, vane, screw, progressing cavity, lobe/com, peristaltic
Positive displacement	Reciprocating	Piston, plunger, diaphragm
Kinetic	Radial flow	Centrifugal
Kinetic	Axial flow	Propeller
Other	Mixed flow	Mixed flow
	Jet/ejector	Jet or ejector type

Affinity laws (same impeller, variable speed):

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2} \quad [-] \quad \frac{H_1}{H_2} = \left(\frac{N_1}{N_2} \right)^2 \quad [-] \quad \frac{P_1}{P_2} = \left(\frac{N_1}{N_2} \right)^3 \quad [-]$$

Speed intuition: double speed $\Rightarrow Q \times 2, H \times 4, P \times 8$.

Affinity laws (same speed, variable diameter):

$$\frac{Q_1}{Q_2} = \frac{D_1}{D_2} \quad [-] \quad \frac{H_1}{H_2} = \left(\frac{D_1}{D_2} \right)^2 \quad [-] \quad \frac{P_1}{P_2} = \left(\frac{D_1}{D_2} \right)^3 \quad [-]$$

Cavitation: suction pressure below vapor pressure $\Rightarrow$ vapor bubbles form/implode. Damages components, especially impeller. Hot fluid, high vapor pressure, high suction losses, low inlet pressure, or pump too high increases risk.

NPSH meaning: net positive suction head = suction-side head margin to avoid cavitation.

NPSH_A available: absolute pressure/head at pump suction port; calculated from the system.

NPSH_R required: minimum suction head required to avoid cavitation; manufacturer/lab tested with water; pump-design specific.

NPSH rule of thumb:

$$NPSH_A \geq 1.1 NPSH_R \quad [\text{m or ft}] \quad \text{else cavitation risk}$$

NPSHA correlation:

$$NPSH_A = H_A \pm H_z - H_f + H_v - H_{vp} \quad [\text{m or ft}]$$

Often: $H_v = v^2/(2g)$ is small, so $NPSH_A \approx H_A \pm H_z - H_f - H_{vp}$.

NPSH terms: H_A absolute pressure head; H_z elevation head, + or -; H_f suction line losses; H_v velocity head; H_{vp} vapor pressure head from fluid tables.

Head from pressure:

$$H = \frac{P}{\gamma} \quad [\text{m or ft}] \quad H[\text{ft}] \approx \frac{2.31 P[\text{psi}]}{SG}$$

NPSH speed effect:

$$(NPSH_R)_2 = \left(\frac{N_2}{N_1} \right)^2 (NPSH_R)_1 \quad [\text{m or ft}]$$

Pneumatics / compressed air: air is compressible; use absolute pressure, absolute temperature, and total volume.

$$PV = mRT \quad [\text{consistent SI or US absolute units}]$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad [\text{absolute units only}]$$

Air capacity rating:

$$Q_1 = Q_2 \left(\frac{P_2}{P_1} \right) \left(\frac{T_1}{T_2} \right) \quad [\text{flow units}]$$

Compressor power form from review:

$$HP = \frac{P_m Q}{654} \left[\left(\frac{P_{out}}{P_{in}} \right)^{0.286} - 1 \right] \quad [\text{hp}]$$

$$KW = \frac{P_m Q}{17.1} \left[\left(\frac{P_{out}}{P_{in}} \right)^{0.286} - 1 \right] \quad [\text{kW}]$$

Mini checklist:

- For real pipe flow: find $Q \rightarrow v \rightarrow Re \rightarrow f \rightarrow h_L$.
- For NPSH: suction side only; use absolute pressure and vapor pressure.
- Check $NPSH_A \geq 1.1 NPSH_R$ after losses and elevation sign.

TABLE 10.5 Friction factor in zone of complete turbulence for new, clean, commercial Schedule 40 steel pipe

Nominal Pipe Size			Nominal Pipe Size		
U.S. (in)	Metric (mm)	Friction factor, f_f	U.S. (in)	Metric (mm)	Friction factor, f_f
1/2	DN 15	0.026	3, 3 1/2	DN 80, DN 90	0.017
3/4	DN 20	0.024	4	DN 100	0.016
1	DN 25	0.022	5, 6	DN 125, DN 150	0.015
1 1/4	DN 32	0.021	8	DN 200	0.014
1 1/2	DN 40	0.020	10-14	DN 250 to DN 350	0.013
2	DN 50	0.019	16-22	DN 400 to DN 550	0.012
2 1/2	DN 65	0.018	24-36	DN 600 to DN 900	0.011